

✅ 📅 DAY 6 — OCR + SCANNED PDF + IMAGE TEXT PIPELINE

🎯 GOAL

Upgrade from: PDF (text only) ✅

to:

PDF (text) ✅

PDF (scanned images) ✅

Images (JPG/PNG) ✅ 🔥

🧠 WHY THIS IS IMPORTANT

Right now your system:

✅ Works for digital PDFs

❌ Fails for scanned PDFs (very common in industry)

❌ Cannot read images

👉 After today:

✅ Your system can see text inside images

✅ ✅ STEP 1 — INSTALL OCR ENGINE

👉 Use: PaddleOCR (best for local use)

✅ Install:

```
pip install paddleocr
```

```
pip install paddlepaddle
```

⚠️ First run will download models

✅ ✅ STEP 2 — CREATE OCR MODULE

📁 ingestion/ocr.py

✅ FULL CODE (WITH EXPLANATION)

```
from paddleocr import PaddleOCR
```

```
class OCRService:
```

```
    def init(self):
```

```
        """    Initialize OCR model    """
```

```
        # Uses English model (change lang if needed later)
```

```
        self.ocr = PaddleOCR(use_angle_cls=True, lang='en')
```

```
    def extract_text(self, image_path):
```

```
        """    Extract text from image    """
```

```
        result = self.ocr.ocr(image_path, cls=True)
```

```
        extracted_text = ""
```

```
        # Loop through detected lines
```

```
        for line in result:
```

```
            for word_info in line:
```

```
                text = word_info[1][0] # actual text
```

```
                extracted_text += text + " "
```

```
        return extracted_text.strip()
```

```
``
```

🧠 WHAT THIS DOES

Image → OCR → Text ✅

✅ STEP 3 — IMAGE INGESTOR

📁 ingestion/image_ingestor.py

✅ CODE

```
from ingestion.ocr import OCRService
from ingestion.cleaning import clean_text
from ingestion.chunking import chunk_text
```

```
def load_image(filepath):
    """    Process image → OCR → chunks    """
    ocr = OCRService()
    # Step 1: extract text
    raw_text = ocr.extract_text(filepath)
    # Step 2: clean
    clean = clean_text(raw_text)
    # Step 3: chunk
    chunks = chunk_text(clean)
    return chunks
```

✅ ✅ STEP 4 — SCANNED PDF SUPPORT

👉 For scanned PDFs, pages are images

We extract page → OCR

📁 ingestion/pdf_ingestor.py (UPDATE)

✅ ADD OCR SUPPORT

```
import pdfplumber

from ingestion.ocr import OCRService
from ingestion.cleaning import clean_text
```

```
from ingestion.chunking import chunk_text
```

```
def load_pdf(filepath):
```

```
    full_text = ""
```

```
    ocr = OCRService()
```

```
    with pdfplumber.open(filepath) as pdf:
```

```
        for i, page in enumerate(pdf.pages):
```

```
            # Try normal text extraction
```

```
            text = page.extract_text()
```

```
            # ✅ If no text → fallback to OCR
```

```
            if not text:
```

```
                print(f"OCR used for page {i+1}")
```

```
                img = page.to_image(resolution=300).original
```

```
                img_path = f"temp_page_{i}.png"
```

```
                img.save(img_path)
```

```
                text = ocr.extract_text(img_path)
```

```
            if text:
```

```
                full_text += f"\n[Page {i+1}]\n"
```

```
                full_text += text
```

```
    cleaned = clean_text(full_text)
```

```
    chunks = chunk_text(cleaned)
```

```
    return chunks
```

🧠 WHAT CHANGED

Before: PDF → text only ❌

Now: PDF → text ✅

if empty → OCR ✅ 🔥

✅✅ STEP 5 — TEST IMAGE INPUT

Put image here:

data/raw/sample.png

✅ CHANGE MAIN TEMPORARILY

```
from ingestion.image_ingestor import load_image
```

```
docs = load_image("data/raw/sample.png")
```

✅ Test:

```
>> what is written in image
```

✅✅ STEP 6 — OPTIONAL ROUTER (SMART INPUT)

📁 app/router.py

```
def route(file_path):
```

```
    if file_path.endswith(".pdf"):
```

```
        return "pdf"
```

```
    elif file_path.endswith(".png") or file_path.endswith(".jpg"):
```

```
        return "image"
```

```
    else:
```

```
        return "text"
```

✅✅ FINAL PIPELINE NOW

Input file

↓

PDF → text OR OCR ✅

Image → OCR ✅

Text → direct 



Cleaning



Chunking



Embeddings



Retrieval + Reranker



LLM answer  

  WHAT YOU CAN NOW DO

Try:

Upload scanned PDF 

Upload screenshot 

Upload invoice image 

Ask:

>> what is in this file

>> summarize content

>> extract values

 SYSTEM LEVEL NOW

You now built:

 Multimodal Document Intelligence System

 CURRENT LIMITATIONS

Feature	Status
Text PDFs	✓
Scanned PDFs	✓
Images	✓
Tables	✗ (Day 7)
Vision reasoning	⚠ basic

🚀 NEXT (DAY 7)

We will add:

- ✓ Table extraction
 - ✓ Excel + PDF comparison
 - ✓ structured data analysis
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